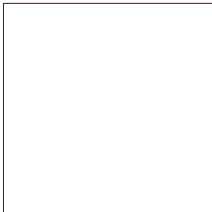


Mixed–Signal Design Course Project PCB

Power



File: Power.kicad_sch

ADC



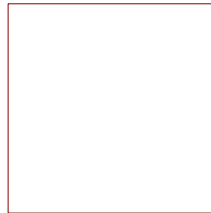
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MCU



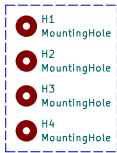
File: MCU.kicad_sch

DAC



File: DAC.kicad_sch

Mounting Holes (M3)



Shrisharva D

Sheet: /
File: Nemesis–MixSigPCB.kicad_sch

Title: Mixed–Signal Design Course Project PCB

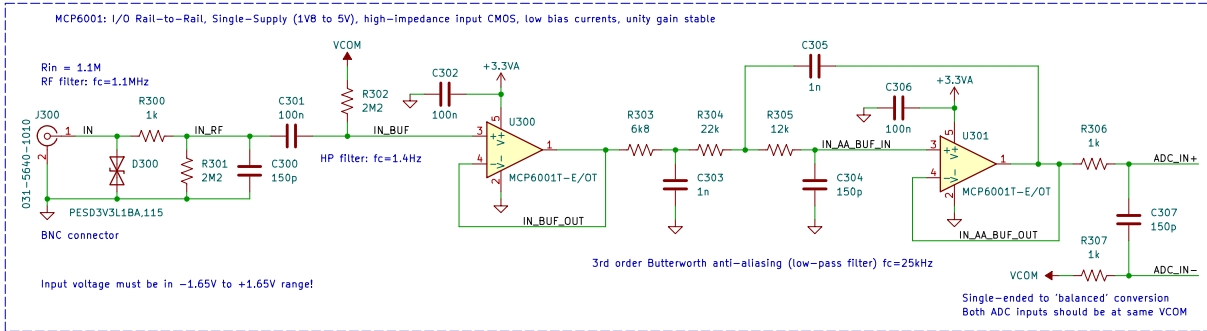
Size: A4 Date: 2026–06–30

KiCad E.D.A. 9.0.7

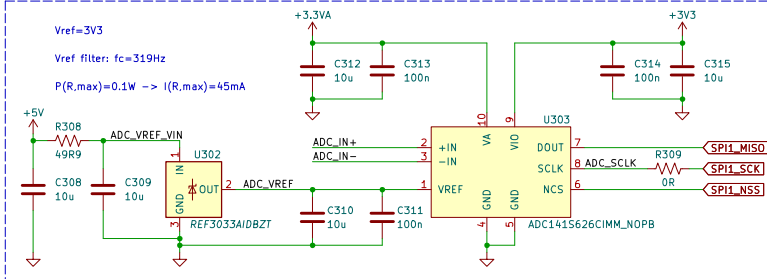
Rev: 1,3

Id: 1/5

Analogue Front-End



14-Bit S/H ADC



Shrisharva D

Sheet: /ADC/
File: ADC.kicad_sch

Title: Mixed-Signal Design Course Project PCB

Size: A4 Date: 2026-06-30

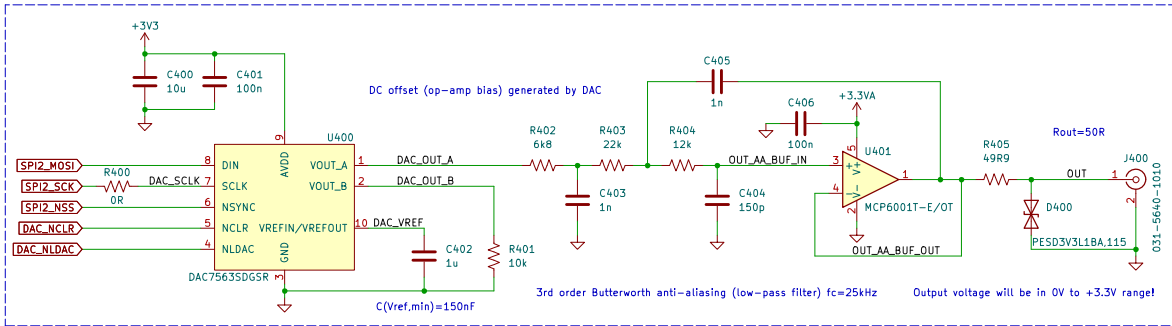
KiCad E.D.A. 9.0.7

Rev: 1.3

Id: 2/5

DAC

Anti-Aliasing Filter and Output Connector (BNC)



Shrisharva D

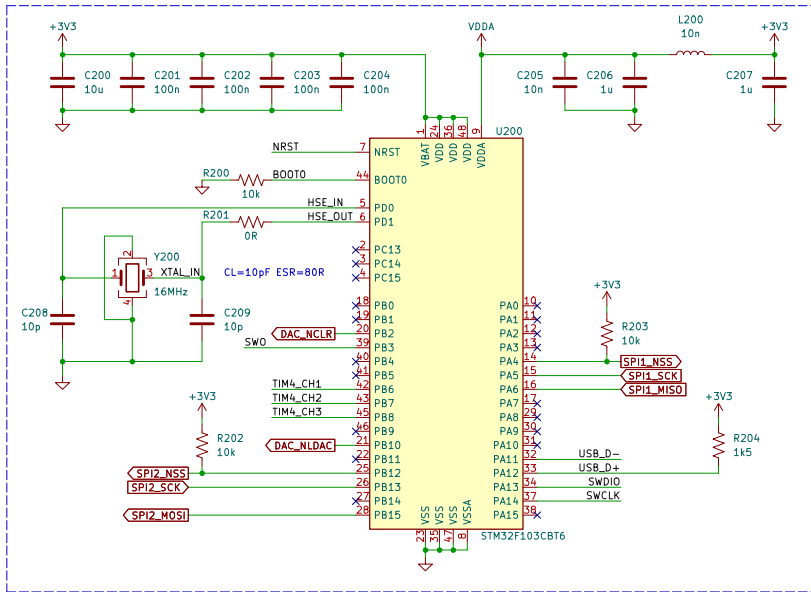
Sheet: /DAC/
File: DAC.kicad_sch

Title: Mixed-Signal Design Course Project PCB

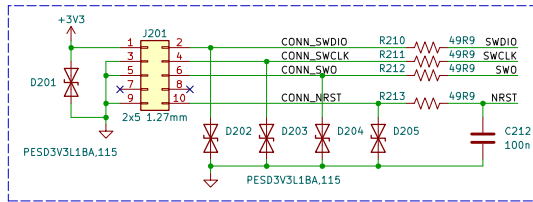
Size: A4 Date: 2026-06-30
KiCad E.D.A. 9.0.7

Rev: 1.3
Id: 3/5

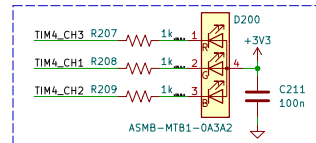
MCU



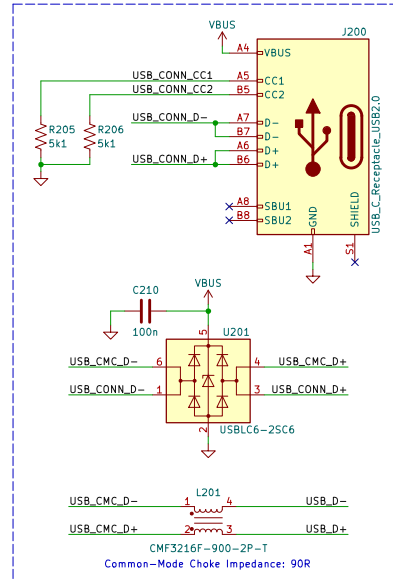
SWD Header



RGB LED



USB Type C



Signal Path:
Connector <-> ESD Protection <-> Common-Mode Choke <-> Pull-Up <-> MCU

Shrisharva D

Sheet: /MCU/
File: MCU.kicad_sch

Title: Mixed-Signal Design Course Project PCB

Size: A4 Date: 2026-06-30

KiCad E.D.A. 9.0.7

Rev: 1.3

Id: 4/5

The schematic diagram illustrates a power supply system for a microcontroller. It features two main voltage regulators: a 5V regulator (U100, HT7533-1) and a 3V3 buck converter (U101, TLV62569DBV). The 5V regulator is powered from a VBUS input through a 10nF capacitor (L100) and a 100nF capacitor (C100). It has a 100k resistor (R100) and a 22uF capacitor (C102) on its output. The 3V3 buck converter is powered from the 5V output through a 15uF capacitor (L101) and a 22uF capacitor (C105). It has a 100k resistor (R101) and a 22k resistor (R102) on its output. The output of the buck converter is connected to a green LED (D100) through a 1k resistor (R103). The schematic also shows various capacitors (C101, C103, C104, C106) and a ground connection.

Rev: 1.3
Id: 5/5